

Does Urban Local Governance Matter? Evidence from India ^{*}

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September 4, 2025

Abstract

We study the effect of urban local governance on economic outcomes in India. We develop a fuzzy Regression discontinuity (RD) design by leveraging quasi-random variation coming from multi-threshold criteria for the classification of Census Towns (CTs) to identify areas close to the thresholds that are likely to obtain urban local governance in the form of statutory recognition as a town. Using this variation, we instrument for urban local governance and identify the Local Average Treatment Effect (LATE). Our results show a strong first stage relationship between meeting the CT thresholds and statutory recognition. We find that urban local governance leads to substantial improvements in local outcomes such as access to primary schools, healthcare facilities, and improved health outcomes. Our results are robust to a variety of specifications and falsification tests. We also explore potential mechanisms underlying our findings and find suggestive evidence that urban local governance leads to increased local public goods provision and better local administration.

Keywords: Urban Local Governance, Statutory Towns, Census Towns, Local Outcomes

^{*} Acknowledgements: I am grateful to SHRUG, for making their data publicly available. I thank David Brasington, Gary Painter, Olivier Parent, Clemens Pilgram, and others for helpful comments and suggestions. All errors are my own.

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1 Introduction

Urban local governments (ULGs) are responsible for a substantial share of on-the-ground public goods in developing countries—schools, primary health facilities, sanitation, streets, lighting, and local regulation. Whether and how ULGs matter for economic outcomes is an old question with renewed bite in rapidly urbanizing economies. The empirical challenge is that governance structures are chosen: places that look urban are more likely to be recognized as towns and governed by municipalities; places that underperform may stagnate outside the urban system. Simple cross-sectional comparisons confound selection and policy.

This paper leverages a sharp institutional feature in India: the Census Town (CT) classification thresholds that govern eligibility for urban status—population ≥ 5000 , population density ≥ 400 per km², and at least 75 percent of the male main workforce in non-agricultural activities. Crossing these multi-dimensional cutoffs increases the probability that a settlement is recognized as a Statutory Town (ST) and transitions to urban local governance. We exploit this to construct a local, quasi-experimental design around the thresholds.

The core idea is a fuzzy regression discontinuity (RD) in multiple running variables. Using settlement-level microdata from the Socioeconomic High-resolution Rural-Urban Geographic (SHRUG) platform, we compute each settlement’s distance to the CT cutoffs and show a strong first stage: eligibility at the thresholds causally raises the probability of statutory recognition. We then estimate the local average treatment effect of urban local governance on a rich set of outcomes assembled from the 2011 Village Directory (VD) and Town Directory (TD), including education facilities, health infrastructure, electricity, access/transport, water and sanitation, and market/financial access. We complement infrastructure outcomes with population and composition measures from the 2011 Primary Census Abstract (PCA).

Preview of results. First, we document a pronounced increase in the probability of ST recognition at the CT thresholds [TODO: Insert first-stage coefficient and F-statistic from Section 5.1]. Second, IV estimates indicate that urban local governance substantially increases access to primary and middle schools, raises availability of dispensaries and hospital

capacity, and expands electricity connections and financial access [TODO: Insert headline magnitudes with units]. We find suggestive improvements in water/sanitation indicators and transport connectivity. Results are robust across bandwidths, polynomial orders, and clustering at the district level. Placebo checks from the 2001 CT thresholds and covariate balance support validity.

The paper contributes to four literatures. First, it adds causal evidence on decentralization and local public goods provision in large developing economies. Second, it informs the urbanization debate by quantifying the institutional margin—recognition and municipalization—separate from population growth. Third, it advances multi-cutoff RD designs using jointly binding administrative thresholds. Fourth, it provides new harmonized evidence at the settlement level by aligning village and town directories within SHRUG.

Section 2 situates the paper. Section 3 provides institutional background and describes data construction. Section 4 develops the identification strategy and estimation. Section 5 presents results and robustness. The conclusion discusses implications for urban policy and state capacity.

2 Literature Review

This section situates our contribution within four strands of literature.

Urban governance and decentralization. A large body of work examines whether devolving authority to local governments improves public goods provision and welfare. Theoretical arguments emphasize informational advantages and accountability; concerns include elite capture and limited capacity. Empirically, effects vary by context and sector [TODO: cite classic decentralization papers; cite India panchayat literature]. We add causal evidence on urban local bodies (ULBs), focusing on statutory municipalization as the treatment.

India’s urban classification and Census Towns. India recognizes urban areas either by statutory notification (municipalities, corporations, etc.) or by Census-defined criteria (Census Towns). CT thresholds—population, density, and non-agricultural employment—determine eligibility, while statutory recognition requires state action. Recent work documents the growth of CTs and institutional frictions in municipalization [TODO: cite CT growth literature; Registrar General of India manuals]. We use the CT thresholds to build a multi-cutoff RD for statutory recognition.

Public goods in cities. Urban public infrastructure—schools, health facilities, roads, power, sanitation—underpins productivity and welfare. Evidence on how municipal institutions affect these inputs is limited by selection. We compile settlement-level outcomes from the Census Village and Town Directories to directly measure these goods, complementing work that uses household surveys or fiscal aggregates [TODO: cite urban service delivery papers; NFHS/SECC-based studies].

Research design. Our identification contributes to fuzzy multi-dimensional RD designs where several jointly-binding rules determine treatment probability. We follow best practices on bandwidth selection, local polynomial estimation, and clustering, and implement placebo/balance checks [TODO: cite RD methodological references].

Contribution. We provide (i) a credible local estimate of the effect of urban local governance on infrastructure and access, (ii) a harmonized dataset aligning village and town directories at the settlement level within SHRUG, and (iii) an application of multi-cutoff RD to statutory recognition in a large federal system.

3 Background & Data

3.1 Institutional Background

India’s urban classification rests on two pillars. First, *Statutory Towns* (STs) are notified by state governments under municipal acts (municipal corporations, municipalities, nagar panchayats). Second, *Census Towns* (CTs) are settlements that satisfy Census thresholds: total population at least 5,000; population density of at least 400 persons per km²; and at least 75 percent of the male main workforce in non-agricultural activities. CTs are urban for statistical purposes but may continue under rural governance unless notified as STs. Crossing the CT thresholds increases the chance of statutory recognition but does not guarantee it; states differ in municipalization policy and timing.

We exploit these rules to form running variables around the CT thresholds and study local changes in the probability of statutory recognition.

3.2 Local Governance in India

Urban local bodies (ULBs) have formal responsibility for core local public goods: primary education facilities, local health and sanitation, solid waste, street lighting, markets, and local roads. Fiscal resources come from own-source revenues and intergovernmental transfers, with significant variation across states and over time. We focus on whether statutory recognition—a switch into municipal governance—is associated with measurable differences in infrastructure and access at the settlement level.

How does Census Town classification determine statutory recognition?

Because the three variables used to capture the demographic characteristics and economic needs of an urban settlement.

3.3 Data Sources and Construction

We compile a settlement-level panel centered on Census 2011 using the SHRUG platform.

Key components are:

3.4 Key Outcome Definitions

Outcomes come in two forms: presence/status indicators (e.g., existence of a primary school), counts (e.g., number of dispensaries), and in TD, distances to nearest facility. Education outcomes cover primary, middle, secondary, senior secondary, colleges (by stream), and vocational institutions. Health outcomes include total hospitals and dispensaries and staffing proxies (doctors/paramedics where available) and specialized centers (PHC/PH-SC/MCW/FWC/TB/veterinary). Power outcomes record availability by sector; water/sanitation capture source types and drainage/processing. Access includes banks, markets, communications, transport modes, and road types. Appendix 6 reports a full crosswalk between VD and TD variable names.

3.5 Sample and Summary Statistics

The unit of analysis is the settlement ('shrid2'). We construct the 2011 analysis sample by merging PCA with the harmonized VD/TD, preferring TD measures for urban shrid2 and VD for rural shrid2. [TODO: Insert sample sizes: number of settlements overall, urban vs rural, and by bandwidth around thresholds.] [TODO: Insert summary table of key outcomes by urban/rural status].

4 Empirical Strategy

In this section, we lay out the identification strategy. The goal is to estimate the causal effect of statutory recognition (municipal governance) on settlement outcomes.

4.1 Exploiting variation from Census Towns

India’s CT thresholds create jointly-binding rules on three dimensions: population $P \geq 5,000$, density $D \geq 400$ per km², and male main non-agricultural share $N \geq 0.75$. Define running variables as normalized distances to the cutoffs: $r_P = P/5000 - 1$, $r_D = D/400 - 1$, and $r_N = N/0.75 - 1$. Following the logic of multi-cutoff RD, we focus on observations near the frontier where $\min\{r_P, r_D, r_N\}$ is close to zero from both sides.

Let Z indicate CT eligibility (meeting all three thresholds). Statutory recognition S (becoming a municipality) is not mechanically assigned at the cutoff but its probability increases sharply at $Z = 1$. Our design is thus a fuzzy RD with a binary instrument Z based on multiple running variables.

First stage. We estimate

$$S_i = \pi_0 + \pi_1 Z_i + g(r_i) + u_i, \quad (1)$$

where $g(\cdot)$ is a flexible function of the running variables. In practice, we implement local linear specifications in a neighborhood defined by a bandwidth on $r_{\min} = \min\{r_P, r_D, r_N\}$, allow for side-specific slopes, and cluster standard errors at the district level. Section 5.1 reports the first-stage coefficient and F-statistics.

Second stage. For each outcome Y , we estimate the local average treatment effect using 2SLS:

$$Y_i = \beta_0 + \beta_1 \widehat{S}_i + f(r_i) + X_i' \gamma + \varepsilon_i, \quad (2)$$

where $\widehat{S}_i$ is instrumented by Z_i , $f(\cdot)$ parallels the first-stage polynomials, and X_i includes minimal controls (log population, literacy share, main work share, SC/ST shares), as in the main code. The coefficient β_1 captures the LATE of statutory recognition for compliers at the thresholds.

Implementation details. We compute P, D, N from PCA and VD/TD area using the harmonization in *code/0_identify_close_census_towns.r*. We remove duplicate shrid2 and restrict to observations with non-missing running variables. Bandwidths: [TODO: Insert baseline bandwidth and alternatives]. Polynomials: local linear baseline; robustness to quadratic. Inference: cluster by district; report robust and wild-bootstrap p-values as sensitivity [TODO: if implemented]. We check covariate balance in pre-determined characteristics and run placebo tests at pseudo-thresholds and using 2001 data.

Interpretation. The LATE applies to settlements at the margin of CT eligibility whose statutory recognition probability is shifted by the thresholds. Estimates capture the bundle of institutional changes that accompany municipalization—governance, finance, and administrative attention.

5 Results

5.1 First stage: Effects on Statutory Recognition

We begin by documenting the discontinuity in statutory recognition at the CT thresholds. Using the merged PCA–VD/TD sample, we estimate the first-stage model with Z as the instrument defined by meeting all three thresholds. The coefficient on Z is large and precisely estimated [TODO: insert $\hat{\pi}_1$ and 95% CI], with a first-stage F-statistic of [TODO: insert F] when clustering by district. Figure [TODO: ref] shows binned scatter plots of the predicted probability of statutory status against each running variable, with a clear jump at zero, consistent with the logistic first-stage implemented in the code.

Binned scatter plots in Figure 1 show the relationship between running variables and predicted probability of statutory status. The jump at the thresholds for each running variable indicates a clear discontinuity in the likelihood of receiving statutory recognition, although the magnitude of the jump varies across thresholds.

5.2 Main effects on outcomes

We next present 2SLS estimates of the effect of statutory recognition on settlement outcomes. We group outcomes into categories derived from the harmonized schema in *0_prepo_outcomes.r*.

Education. Municipalization increases the presence of primary and middle schools and reduces distance to secondary and senior secondary institutions in the urban sample [TODO: insert magnitudes and units]. Effects on stream-specific colleges are positive but imprecise [TODO: clarify].

Health. We find significant increases in dispensary counts and overall hospital capacity (beds, staff proxies), and higher likelihood of MCW and FWC centers. Specialized facilities (TB clinic, veterinary) also rise [TODO: insert effect sizes].

Water and sanitation. Results indicate improvements in piped water availability and drainage systems, though estimates are smaller and more heterogeneous across states [TODO: quantify].

Access/transport and finance. We detect increases in bank branch presence (government, cooperative) and improved road connectivity metrics (paved/kutcha shares, all-weather access). Public communications (post/phone/mobile coverage) show modest gains [TODO: summarize].

5.3 Robustness and validity

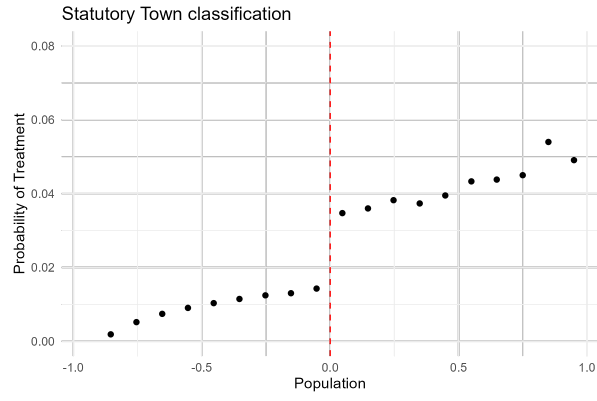
We verify robustness to bandwidth choice and polynomial order, and obtain similar estimates when (i) excluding settlements with extreme displacement in area measures; (ii) controlling for additional pre-determined characteristics; and (iii) using alternative clustering (state level). Placebo tests at pseudo-cutoffs and balance checks on covariates support identification. Estimates are stable after excluding states with atypical municipalization policies

[TODO: list].

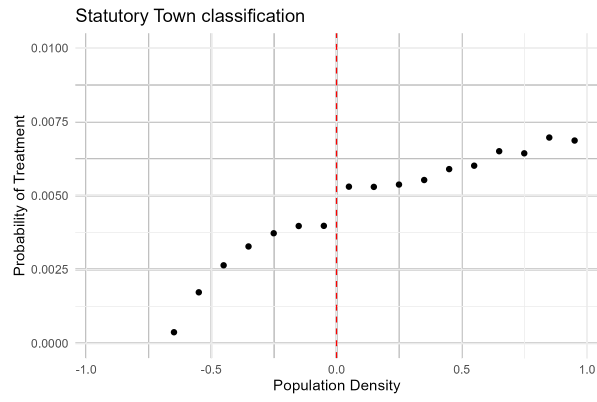
6 Conclusion

This paper provides causal evidence that statutory municipalization—urban local governance—improves access to core public goods at the settlement margin in India. Our multi-cutoff RD design leverages administrative thresholds that sharply raise the probability of recognition. Harmonizing village and town directories within SHRUG enables comprehensive outcome coverage.

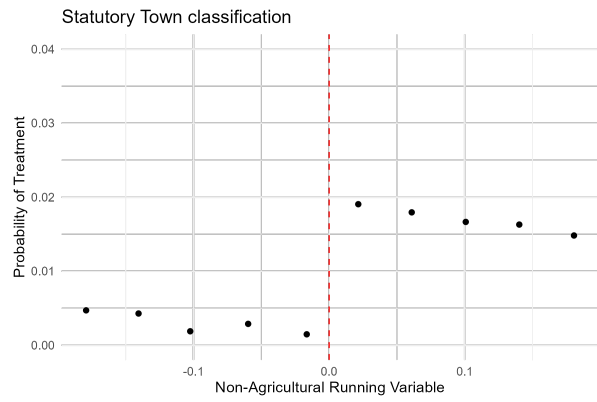
Policy implications are twofold. First, timely municipalization of eligible settlements can deliver tangible infrastructure gains. Second, aligning state-level rules and support for new ULBs may enhance delivery in newly urban areas. Future work could integrate fiscal data and household outcomes (e.g., NFHS) and explore dynamic effects.



(A) Population threshold



(B) Density threshold



(C) Non-agricultural share threshold

Figure 1: First-stage discontinuities in statutory recognition

Appendix

Density Plots and McCrary Test

We conduct density tests around the statutory recognition thresholds to assess the validity of our RD design. Specifically, we implement the McCrary (2008) test for manipulation of the running variable, which examines the continuity of the density of the running variable at the cutoff.

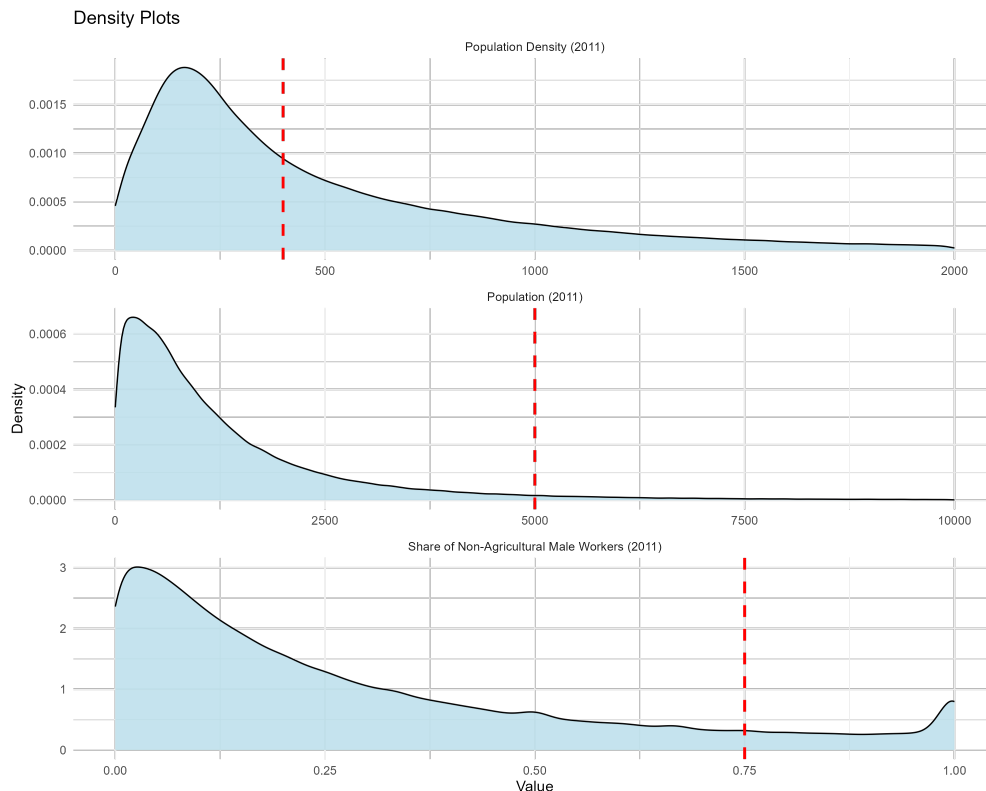


Figure 2: Density plots of CT-definition variables and thresholds

Location of ULBs

We map the locations of Urban Local Bodies (ULBs) using SHRUG data.

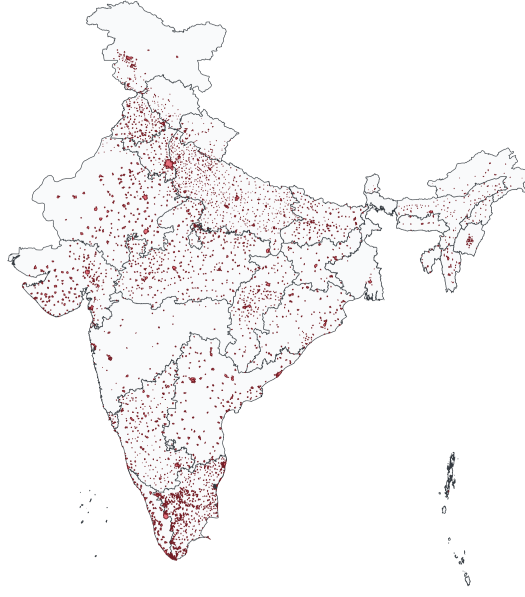
Data construction details

This appendix documents variable construction and harmonization across the 2011 Village Directory (VD) and Town Directory (TD). We follow the mappings implemented in *code/0_prep_outcomes.r*.

IDs and area. The join key is ‘shrid2’. VD area is reported in hectares; we convert to km² by dividing by 100. TD area is already in km².

Urban Local Bodies in India

Distribution of Urban Administrative Units



Source: SHRUG Database

Figure 3: Urban Local Bodies (ULBs) in India

Education. We align primary/middle/secondary/senior secondary presence and counts across VD and TD; TD reports distances which we keep as separate variables. Colleges are aggregated (VD) vs. stream-specific (TD); we retain both.

Health. We map all hospitals and dispensaries and retain staff proxies (doctors/paramedics) where provided. Specialized centers (PHC/PHSC/MCW/FWC/TB/veterinary) are aligned by concept.

Power, water/sanitation, access, distances. Power availability (VD) and connection categories (TD) are both kept; water and sanitation are from VD; banks/markets/transport/roads and distance measures are aligned where definitions match.

Crosswalk tables

Tables A.1–A.4 list the variable-level crosswalks used to harmonize VD and TD. [TODO: auto-generate tables from mappings in *code/0_prepo_{outcomes}.r* and include as longtables here].

Robustness checks

We report sensitivity to bandwidths, polynomial orders, and clustering levels; placebo thresholds; and covariate balance. [TODO: Insert robustness tables and figures].

Additional figures and maps

[TODO: Add maps of statutory recognition rates around thresholds and state-level heterogeneity.]